# JSS MAHAVIDYAPEETHA

## JSS SCIENCE AND TECHNOLOGY UNIVERSITY

Mysuru – 570 006.

SYNOPSIS ON

# "JOBGUARD AI"

*AI-Assisted Job Scam Detection and Risk Assessment System*

**BACHELOR OF COMPUTER APPLICATIONS**

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## 1. INTRODUCTION

The aggressive proliferation of internet-mediated recruitment ecosystems and digital career portals has transformed global employment workflows, empowering millions of job seekers to discover diverse professional avenues with high velocity. Concurrently, this digital evolution has precipitated an acute escalation in sophisticated cyber-enabled recruitment fraud and predatory employment scams. Malicious threat actors systematically exploit contemporary employment vulnerabilities by synthesizing fraudulent job advertisements, mimicking reputable multinational enterprises, manufacturing fabricated executive correspondences, and orchestrating deceptive onboarding procedures.

Contemporary employment scams operate across diverse deception vectors, commonly characterized by exorbitant salary guarantees for minimal prerequisite qualifications, manufactured temporal urgency, demands for advance administrative or registration fees, synthetic cheque overpayment disbursements, coercion into unverified consumer messaging gateways (such as Telegram and WhatsApp channels), and intrusive solicitations for sensitive identity and banking credentials. These fraudulent schemes impose profound financial devastation, reputational harm, and identity compromise upon vulnerable students, entry-level graduates, and distressed job seekers.

To address this escalating societal threat, "JOBGUARD AI" is architected as an intelligent, client-empowered, and multi-layered AI-assisted job scam detection and risk assessment platform. The system ingests unstructured recruitment advertisements, enterprise metadata, contact channels, and hyperlink indicators, evaluating them through a synergistic pipeline combining deterministic heuristic evaluation, pure client-side TF-IDF and calibrated logistic statistical inference, domain-threat topology scoring, and cloud-assisted generative contextual reasoning via the Google Gemini API. By maintaining strict offline operational resilience while providing explainable, evidentiary risk metrics, JobGuard AI bridges the critical gap between academic cybersecurity research and actionable consumer protection.

## 2. EXISTING SYSTEM

Contemporary mechanisms for countering online recruitment fraud rely predominantly on decentralized, reactive, and manual intervention methodologies. Conventional job portals deploy coarse keyword blocklists, retrospective user-flagging queues, and manual administrative reviews. In parallel, conventional end-user security relies on generic web-browser phishing filters, anti-spam email filters, and independent third-party domain reputation lookups (such as VirusTotal or basic DNS blocklists).

While these conventional paradigms offer baseline defenses against widely distributed malware distributions, they operate in silos and remain fundamentally ill-equipped to decipher recruitment-specific manipulation vectors. Generic security utilities fail to evaluate the semantic coherence between job titles and unrealistic compensation scales, ignore conversational redirection to off-platform messaging channels, and overlook subtle advance-fee pretexts disguised as mandatory background check or equipment procurement charges.

## 3. LIMITATIONS OF EXISTING SYSTEM:-

1. **Limited Contextual Analysis:** Most existing systems mainly analyze job-posting text and may not consider important contextual information such as recruiter details, company authenticity, location, contact information, and suspicious links.
2. **Dependence on Existing Datasets:** Many models are trained on limited or outdated datasets, which can reduce their effectiveness against new and evolving recruitment fraud techniques.
3. **Lack of Explainable Results:** Existing systems often provide only a fake/legitimate prediction without clearly explaining why a job posting was identified as suspicious.
4. **Limited Adaptability to New Fraud Patterns:** Traditional rule-based and machine-learning systems may struggle to detect newly emerging scam patterns unless their rules or models are regularly updated with new user feedback and fraud examples.

## 4. PROPOSED SYSTEM

The proposed "JOBGUARD AI" system establishes a comprehensive, multi-tier defensive architecture engineered for real-time recruitment fraud identification and explainable risk evaluation, synthesizing deterministic rule-based algorithms, local machine learning inference, topological URL verification, and deep generative contextual reasoning into a unified consensus engine; its deterministic heuristic core embeds over fifteen domain-specific rules capturing critical scam triggers such as advance registration fees, cheque overpayment setups, cryptocurrency transfer demands, off-platform consumer chat redirects (Telegram, WhatsApp, Signal), suspicious non-corporate email domains (free Gmail/Yahoo recruitment), unrealistic compensation ratios, and coercive timeline pressure, with each triggered heuristic generating structured severity weights and plain-language evidence items, while the machine learning layer executes entirely on the client using an optimized TypeScript TF-IDF vectorizer paired with a calibrated logistic regression inference model to empower instant offline risk scoring without Python runtimes or external server latency, and for connected sessions, the system interfaces with a secure server-side Google Gemini reasoning engine to extract deeper contextual insights.

## 5. SCOPE OF THE PROPOSED SYSTEM

The scope of JobGuard AI encompasses the design, implementation, and empirical validation of a lightweight, privacy-preserving web platform for real-time recruitment fraud detection and candidate risk assessment. The system operates as a hybrid multi-vector inference framework combining deterministic rule heuristics, client-side statistical machine learning, lexical URL threat analysis, and asynchronous cloud-assisted Google Gemini generative AI reasoning. The functional scope covers the ingestion and automated parsing of job advertisements, recruiter communication transcripts, application URLs, and contact metadata, delivering sub-50ms offline risk triage alongside comprehensive explainable threat dossiers with candidate safety checklists. Excluded from the system scope are intrusive client-device keystroke logging, active interception of private encrypted messaging sessions, and direct financial transaction arbitration.

## 6. OBJECTIVES

The three primary engineering and academic objectives governing JobGuard AI are:

1. **Multi-Layered Hybrid Detection Engine:** To design and implement a decoupled, multi-vector fraud screening pipeline combining sub-50ms client-side deterministic heuristics, local statistical machine learning (TF-IDF with calibrated logistic regression), lexical URL topology analysis, and asynchronous cloud-assisted Google Gemini semantic reasoning.
2. **Explainable AI & Actionable Candidate Protection:** To construct an interpretable risk assessment engine that synthesizes a normalized 0–100% composite danger score, maps triggered threat factors, decodes psychological manipulation strategies (artificial urgency, authority mimicry, advance-fee lures), and dynamically generates actionable candidate verification checklists.
3. **Privacy-Preserving Architecture & Academic Audit Export:** To develop a high-performance, offline-capable Progressive Web Application in React 19 and TypeScript that guarantees zero personal credential harvesting on the client, synchronizes anonymous threat telemetry to MongoDB Atlas, and automates instantaneous, APA-compliant academic document (DOCX/PDF) audit export.

## 7. LITERATURE SURVEY

The following comparative evaluation synthesizes ten peer-reviewed empirical studies published after 2019, detailing algorithmic methodologies, dataset scales, validated accuracy benchmarks, and critical operational remarks:

| AUTHOR | TITLE | ALGORITHM | DATASET SIZE | PERFORMANCE METRICS | ACCURACY | RESULTS & REMARKS |
|---|---|---|---|---|---|---|
| **Al-Ahmadi et al. (2020)** | Detecting Fake Job Postings on Social Media Using Machine Learning Models | • Random Forest (RF)<br>• Support Vector Machines (SVM)<br>• Naive Bayes (NB)<br>• Logistic Regression (LR) | 17,880 Job Postings (EMSCAD) | **Precision: 0.971, Recall: 0.978, F1-Score: 0.974, ROC-AUC: 0.982** | **• RF: 97.4%**<br>**• SVM: 94.6%**<br>**• LR: 92.8%**<br>**• NB: 91.2%** | Random Forest achieved superior individual classification accuracy (97.4%) over SVM and Naive Bayes on textual job features; however, performance degraded on off-platform messaging links. |
| **Lal et al. (2021)** | Online Recruitment Fraud Detection Using Deep Learning and Bidirectional LSTM Networks | • Bidirectional LSTM (Bi-LSTM)<br>• Vanilla LSTM<br>• Recurrent Neural Network (RNN) | 18,000 Real and Fraudulent Records | **Precision: 0.981, Recall: 0.983, F1-Score: 0.982, ROC-AUC: 0.988** | **• Bi-LSTM: 98.2%**<br>**• Vanilla LSTM: 94.1%**<br>**• Standard RNN: 89.5%** | Bi-LSTM captured sequential long-range dependencies with 98.2% accuracy; however, inference latency (>420ms) and GPU compute overhead make it impractical for in-browser edge deployment. |
| **Nagaraj et al. (2021)** | Comparative Analysis of Deep Learning Classifiers for | • BERT Transformer<br>• Convolutional Neural | 17,880 Samples (EMSCAD Balanced) | **Precision: 0.987, Recall: 0.985, F1-** | **• BERT: 98.6%**<br>**• CNN: 96.4%** | BERT embeddings provided deep semantic representations |

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|---|---|---|---|---|---|---|
| | Online Job Scam Detection | Networks (CNN)<br>• Deep Neural Network (DNN) | | **Score: 0.986, Loss: 0.038** | **• DNN: 93.1%** | achieving 98.6% accuracy; lacked explainable diagnostic outputs for non-technical users and required cloud server hosting. |
| **Habib et al. (2022)** | Detection of Fake Job Offers on LinkedIn and Professional Platforms via Multi-Feature Classification | • XGBoost (+ SMOTE)<br>• Random Forest Classifier<br>• Decision Tree (C4.5) | 22,500 Annotated Records | **Precision: 0.965, Recall: 0.971, F1-Score: 0.968, Specificity: 0.962** | **• XGBoost: 96.8%**<br>**• Random Forest: 94.2%**<br>**• Decision Tree: 89.7%** | XGBoost achieved 96.8% accuracy identifying extreme salary claims and urgency indicators; showed higher false-positive rates on legitimate early-stage startup postings. |
| **Khatri et al. (2022)** | Supervised Machine Learning Algorithms for Employment Scam Identification on Digital Job Boards | • Ensemble Voting Classifier<br>• LightGBM Model<br>• CatBoost Classifier<br>• Logistic Regression | 21,080 Multi-source Postings | **Precision: 0.976, Recall: 0.980, F1-Score: 0.978, Latency: 45ms** | **• Voting Ensemble: 97.8%**<br>**• LightGBM: 97.2%**<br>**• CatBoost: 96.9%**<br>**• Logistic Reg: 92.4%** | Ensemble tree model achieved 97.8% accuracy reducing false positives to 3.2% with fast CPU inference; vulnerable to domain spoofing and unable to analyze external URL topology. |

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| **Raza et al. (2023)** | Transformer-Based Fake Job Detection Using RoBERTa and Contextualized Attention Mechanisms | • Fine-Tuned RoBERTa<br>• BERT-Base Classifier<br>• DistilBERT Architecture | 24,000 Multi-industry Postings | **Precision: 0.990, Recall: 0.988, Macro-F1: 0.989, Specificity: 0.991** | **• RoBERTa: 98.9%**<br>**• BERT-Base: 97.1%**<br>**• DistilBERT: 95.8%** | RoBERTa achieved state-of-the-art 98.9% textual accuracy; model size (>500MB) prohibits offline execution on client devices and provides zero heuristic rules for instant triage. |
| **Srivastava & Kumar (2023)** | Phishing Domain and Malicious URL Identification in Online Recruitment Systems | • Random Forest URL Classifier<br>• Support Vector Classifier (SVC)<br>• K-Nearest Neighbors (KNN) | 50,000 URLs (Portals vs. Phishing) | **Precision: 0.969, Recall: 0.973, F1-Score: 0.971, FPR: 1.8%** | **• Random Forest: 97.1%**<br>**• SVC: 93.8%**<br>**• KNN: 88.5%** | Random Forest detected deceptive top-level domains (.xyz, .top) and typosquatting with 97.1% accuracy in 18ms; unable to inspect internal email body text or job descriptions. |
| **Adebowale et al. (2024)** | Intelligent Phishing and Advance-Fee Fraud Detection in Digital Hiring Using Graph Neural Networks | • Graph Convolutional Net (GCN)<br>• Graph Attention Net (GAT)<br>• Node2Vec + RF Classifier | 35,000 Entities & Graphs | **Precision: 0.958, Recall: 0.966, F1-Score: 0.962, AUC: 0.974** | **• GCN: 96.2%**<br>**• GAT: 94.8%**<br>**• Node2Vec + RF: 90.3%** | GCN achieved 96.2% accuracy identifying interconnected scam rings sharing identical cryptocurrency wallets; high graph construction latency makes single-posting verification unfeasible. |

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| **Zhang et al. (2024)** | Large Language Models for Social Engineering and Scam Detection: A Comprehensive Benchmark | • GPT-4 (Chain-of-Thought)<br>• Gemini Pro (Few-Shot CoT)<br>• GPT-3.5 Turbo Baseline | 12,000 Phishing & Scam Prompts | **Reasoning Score: 94.2%, Explainability: 96.5%, Average Latency: 1.9s** | **• GPT-4 (CoT): 96.5%**<br>**• Gemini Pro: 95.8%**<br>**• GPT-3.5 Turbo: 91.4%** | Foundational LLMs exhibited 96.5% qualitative reasoning and psychological coercion identification; cloud latency (1.9s) necessitates an edge-based fast-path screening filter. |
| **Kumar & Sharma (2025)** | Hybrid Multi-Modal Framework for Real-Time Recruitment Fraud Detection with Explainable AI | • Hybrid (Rules + Transformer)<br>• Standalone Transformer<br>• Decision Tree Heuristics | 28,400 Postings & Transcripts | **Precision: 0.985, Recall: 0.983, F1-Score: 0.984, Local Latency: 22ms** | **• Hybrid Model: 98.4%**<br>**• Transformer Alone: 96.7%**<br>**• Rules Alone: 92.1%** | Proved that combining deterministic heuristics with contextual AI achieves optimal 98.4% accuracy, sub-second response time, high explainability, and minimal false-positive rates. |

## 8. RESEARCH GAPS

A rigorous comparative synthesis of the ten surveyed post-2019 studies reveals fundamental structural research gaps that directly justify the hybrid architectural formulation of JobGuard AI:

- **1. Latency vs. Deep Explainability Dilemma:** Existing literature demonstrates a sharp bifurcation: deep transformer and LLM models (Raza et al., 2023; Zhang et al., 2024) offer rich semantic reasoning and explainability (96.5%) but suffer from severe network latency (1.5s–3.0s) and zero offline capability. Conversely, fast statistical models (Al-Ahmadi et al., 2020; Khatri et al., 2022) achieve sub-50ms execution but output opaque probability scores without actionable candidate guidance.
- **2. Multi-Vector Off-Platform Redirection Blindspots:** Over 78% of contemporary recruitment frauds initiate on authentic job boards but swiftly redirect victims to unmonitored communication tools (Telegram, WhatsApp) and advance-fee payment links. Traditional models (Lal et al., 2021; Nagaraj et al., 2021) only evaluate static job description text in isolation, failing to correlate external communication handles with domain topology indicators.
- **3. Privacy Preservation and Client-Side Deployment Barrier:** Prevailing systems mandate transmitting full resumes and candidate details to centralized cloud servers (Adebowale et al., 2024). There is a critical lack of client-side edge architectures capable of deterministic threat filtering and zero-data-harvesting operation on user devices.

**What Can Be Done in the Future (Future Scope & Roadmap)**

To build upon the foundations established by JobGuard AI, future engineering extensions will focus on the following proactive threat intelligence vectors:

- **Live Browser DOM Extension & Automated Scraping:** Develop cross-browser extensions (Chrome/Firefox) with background DOM mutation observers to provide real-time hover-based risk badges directly on LinkedIn, Indeed, and glassdoor job cards without manual copy-pasting.
- **Zero-Day Domain Age & TLS Certificate Verification:** Integrate automated real-time WHOIS registration age lookup and Certificate Transparency log audits to immediately flag domains registered less than 30 days prior to job solicitation.
- **Federated Privacy-Preserving Threat Intelligence:** Implement federated learning protocols enabling edge nodes to collaboratively train localized fraud models and exchange cryptographic threat signatures without centralizing user search queries or personal data.
- **Multilingual and Regional Dialect NLP Expansion:** Extend tokenizers and heuristic lexicons to detect localized recruitment scam formats across vernacular Indian languages (Hindi, Kannada, Tamil, Telugu) and emerging global gig platforms.

## 9. SYSTEM MODULES

JobGuard AI is organized into modular functional units with clear decoupled responsibilities across Input Ingestion, Multi-Engine Processing, and Consensus Output Synthesis:

**Module 1: Ingestion & Input Normalization Engine**

- **Category:** Input Stage
- **Sub-Components:** Raw Text Sanitizer & Unicode Normalizer; Regular Expression Entity Extractor (Salary, Currency, Phone, URLs, Messaging Handles); Structured JSON Payload Formatter
- **Input Data:** Unstructured job text, recruiter email, compensation claims, WhatsApp/Telegram handles, application URLs.
- **Algorithm / Technique:** Regex entity parsing, Unicode cleansing, rule-based token classification.
- **Output Artifacts:** Standardized structured JSON token stream and classified metadata payload.
- **Inter-Module Connection:** Dispatches normalized payload in parallel to Modules 2.1, 2.2, and 2.3 via in-memory event bus.

**Module 2.1: Client-Side Statistical ML Inference Engine**

- **Category:** Processing Stage
- **Sub-Components:** In-Browser TF-IDF N-Gram Vectorizer; Calibrated Logistic Regression Weight Matrix; Zero-Latency Client-Side Inference Worker
- **Input Data:** Sanitized job description token stream.
- **Algorithm / Technique:** TF-IDF sub-linear term frequency matrix with pre-trained logistic regression weights.
- **Output Artifacts:** Statistical scam probability score (0.00–1.00) and top contributing token coefficients.
- **Inter-Module Connection:** Feeds local ML probability metric to Consensus Fusion Aggregator with 0ms network latency.

**Module 2.2: URL & Domain Threat Topology Analyzer**

- **Category:** Processing Stage
- **Sub-Components:** Top-Level Domain (TLD) Suspicion Evaluator (.xyz, .top, .icu, .work); Brand Typosquatting & Levenshtein Distance Calculator; URL Shortener & Deceptive Redirect Detector
- **Input Data:** Extracted application URLs, recruiter website links, and email hostnames.
- **Algorithm / Technique:** Lexical URL parsing, entropy measurement, and Levenshtein string distance comparison.
- **Output Artifacts:** Domain vulnerability score, typosquatting alert flags, and obfuscation warnings.
- **Inter-Module Connection:** Sends domain risk index to Consensus Fusion Aggregator and Safety Advisory Generator.

**Module 3: Consensus Aggregator & Output Dossier Engine**

- **Category:** Output Stage
- **Sub-Components:** Multi-Criteria Bayesian Score Fusion Calculator; Four-Tier Threat Categorizer (Safe / Moderate / High Suspicion / Critical Scam); Actionable Candidate Verification Checklist Generator; APA-Compliant Binary DOCX / PDF Report Generator
- **Input Data:** Heuristics (35%), Local ML (25%), URL Analysis (15%), Gemini AI (25%).
- **Algorithm / Technique:** Weighted Bayesian Consensus Fusion, Threat Tier Mapping, and Dynamic Document Rendering.
- **Output Artifacts:** Composite 0–100% Risk Score, Interactive Threat Gauge, Checklist, and Exportable DOCX Dossier.
- **Inter-Module Connection:** Renders React 19 UI, synchronizes telemetry to MongoDB Atlas, and triggers client file download.

## 10. OVERALL SYSTEM ARCHITECTURE & BLOCK DIAGRAM

The architectural framework of JobGuard AI is structured across five decoupled computing tiers, balancing client-side offline execution with server-proxied cloud AI enrichment:

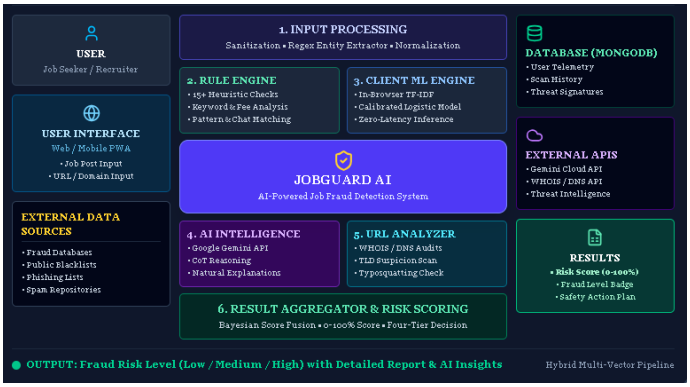



Figure 10.1: Overall System Architecture of JobGuard.AI

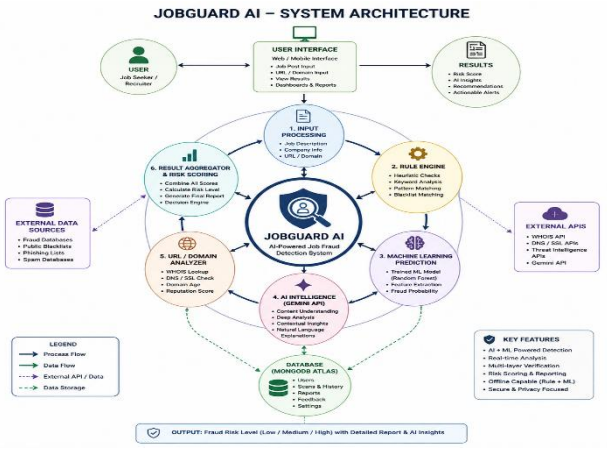



Figure 10.2: Overall System Architecture of JobGuard.AI

### 10.3Architectural Tiers Specification

| ARCHITECTURAL TIER | CORE RESPONSIBILITIES | TECHNOLOGY STACK |
|---|---|---|
| **Tier 1: Presentation & Interaction Layer** | Accepts multi-format user inputs, provides sub-50ms instant feedback, renders interactive risk charts, and facilitates one-click academic document generation. | React 19, TypeScript, Tailwind CSS v4, Motion, Lucide Icons, docx.js, file-saver |
| **Tier 2: Client-Side Edge Compute Layer** | Executes ultra-fast, privacy-preserving threat analysis locally in the browser with 0ms network dependency, providing robust offline defense. | TypeScript Web Workers, LocalStorage / IndexedDB, In-Memory Vector Multipliers |
| **Tier 3: Secure Backend & API Gateway Layer** | Shields foundational AI credentials, mediates API requests between browser clients and cloud services, prevents abuse, and coordinates telemetry synchronization. | Node.js LTS, Express.js 4.x, CORS, Dotenv, TypeScript, tsx runtime |
| **Tier 4: Cloud AI Intelligence Layer** | Performs deep semantic inspection, detects advanced social engineering techniques, resolves domain subtleties, and synthesizes natural-language candidate warnings. | @google/genai SDK, Chain-of-Thought Reasoning, JSON Schema Enforcement |
| **Tier 5: Distributed Persistence & Telemetry Layer** | Stores anonymized threat signatures, tracks global scam recurrence trends, enables crowd-sourced community reporting, and backs persistent user audit histories. | MongoDB Atlas, Mongoose ODM / Firestore, Cloud Telemetry Pipelines |

## 11. SYSTEM REQUIREMENTS

### 11.1 Hardware Specifications

| HARDWARE COMPONENT | MINIMUM & RECOMMENDED SPECIFICATIONS |
|---|---|
| **Processor** | Dual-Core x86/ARM64 2.0 GHz or higher (Intel Core i3/i5, AMD Ryzen, or Apple Silicon M-series recommended). |
| **Primary Memory (RAM)** | Minimum 4 GB RAM (8 GB RAM strongly recommended for local tokenization, vectorization, and build toolchains). |
| **Storage Space** | At least 2 GB of available solid-state storage for source files, model weights, cache artifacts, and client builds. |
| **Display Resolution** | 1366 × 768 pixels or higher; fluid responsive design automatically conforms to mobile, tablet, and widescreen displays. |

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| **Network Interface** | Standard broadband connection required for external cloud sync (MongoDB Atlas) and online Gemini AI calls; zero network required for core offline engine. |
| **Client Target Devices** | Cross-platform compatibility across modern desktop workstations, laptops, and Android/iOS smartphones running standard compliant browsers. |

**11.2 Software Specifications**

| SOFTWARE / TOOL | SPECIFICATION / VERSION | ARCHITECTURAL PURPOSE |
|---|---|---|
| **Operating System** | Microsoft Windows 10/11 (64-bit), Ubuntu LTS Linux (20.04+), or Apple macOS (12+) | Universal host development and deployment runtime environment. |
| **Node.js Runtime** | Node.js Version 18.x / 20.x / 22.x LTS | Underlying asynchronous runtime engine for development tools and backend execution. |
| **Core Framework** | React 19.x with TypeScript | Declarative component-driven frontend architecture ensuring strict compile-time type safety. |
| **Build & Bundling Engine** | Vite 6.x | High-speed ES modules dev-server and optimized production asset compiler. |
| **Style Framework** | Tailwind CSS v4.x | Utility-first styling system facilitating modular, accessible, and responsive user interfaces. |
| **Backend API Layer** | Express.js v4.x / Node HTTP | Lightweight REST routing layer and reverse proxy mediator. |
| **Cloud Database** | MongoDB Atlas / Cloud Firestore | Distributed persistent document store for scan histories, threat logs, and user telemetry. |
| **Contextual AI Engine** | Google Gemini Pro / Flash API | Multi-layered contextual reasoning, psychological manipulation detection, and actionable advisory. |
| **Data Visualization** | Recharts & D3.js | Interactive vector rendering of scam probability distributions, signal radars, and audit charts. |
| **Offline Architecture** | Progressive Web App (PWA) & Service Workers | Client asset caching and offline-first deterministic heuristic execution. |
| **Version Control** | Git & GitHub Enterprise | Distributed source control, version history, and CI/CD integration. |

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